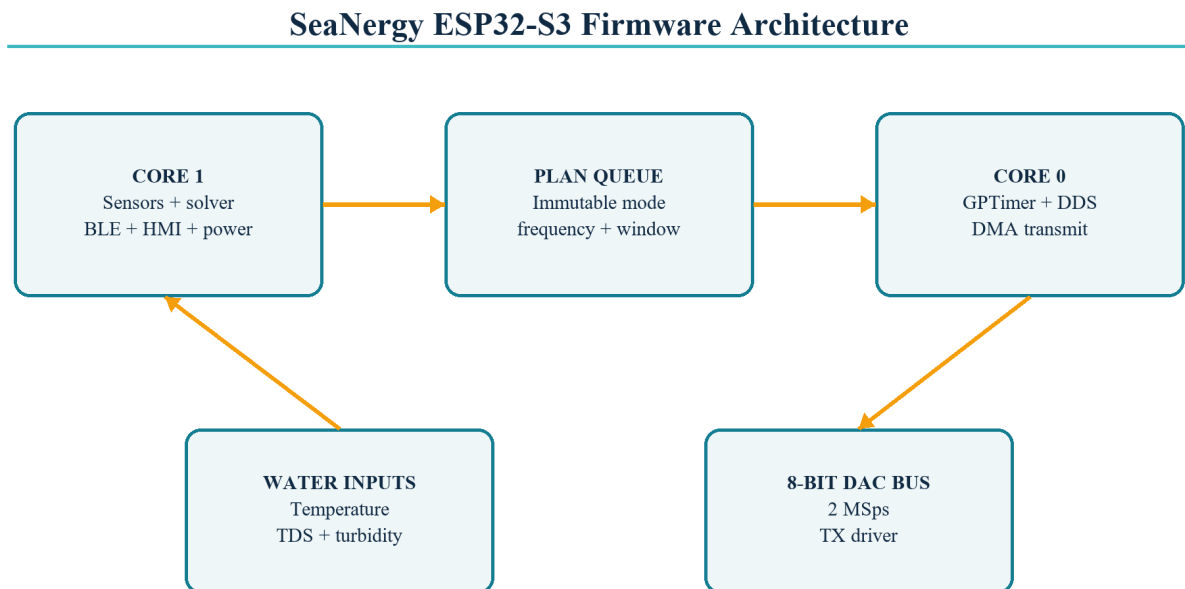


# SeaNergy Payload Firmware

## End-to-End Technical Reference

ESP32-S3 adaptive acoustic payload for SIH 2026 Problem Statement 26058

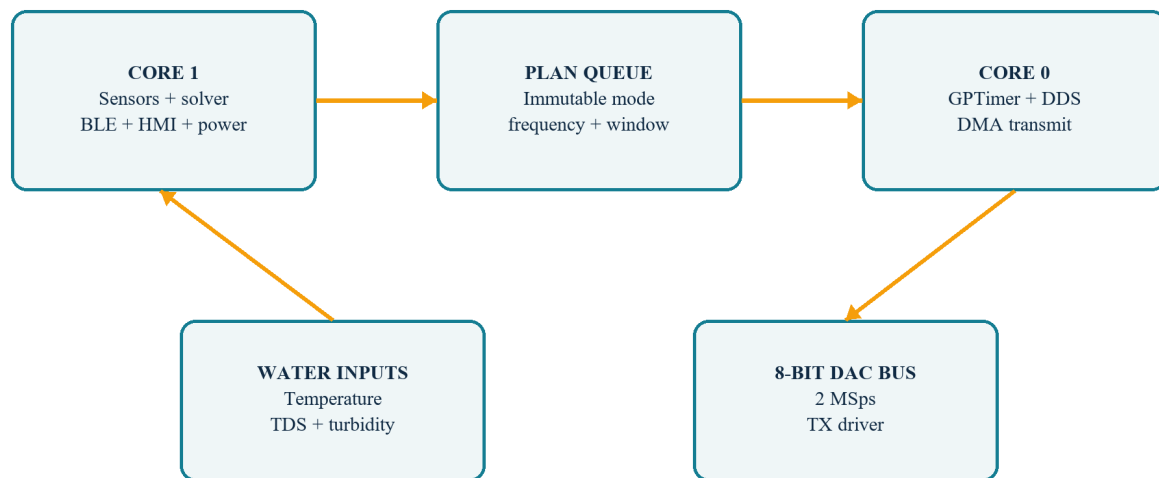


## Release scope

The firmware combines deterministic transmit generation, environmental adaptation, BLE interoperability, local HMI, power accounting and externally measurable CPU-idle behaviour. All packet fields and hardware timing interfaces are frozen in this document.

# 1. Architecture

## SeaNergy ESP32-S3 Firmware Architecture



Core 0 is reserved for the transmit task and its DMA/timer callbacks. Core 1 performs sensing, selection, BLE, HMI and energy accounting. Two static 1,024-byte buffers are allocated in internal DMA-capable SRAM.

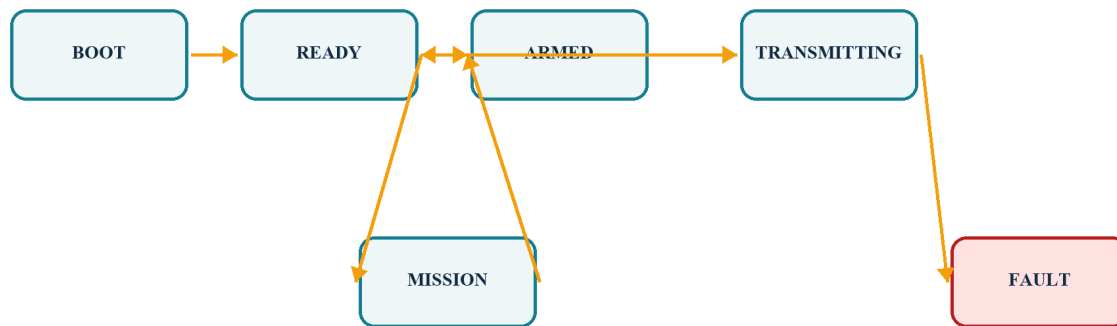
| Executor | Core | Priority | Stack       | Budget              |
|----------|------|----------|-------------|---------------------|
| Transmit | 0    | 20       | 4,096 B     | 2.098 ms range ping |
| Control  | 1    | 8        | 6,144 B     | 1 ms / 20 ms        |
| Sensors  | 1    | 5        | 4,096 B     | 820 ms / 1 s        |
| NimBLE   | 1    | 5        | IDF-managed | 2 ms event          |

## 2. Timing and state control

### Range-Mode Per-Ping Timing Budget

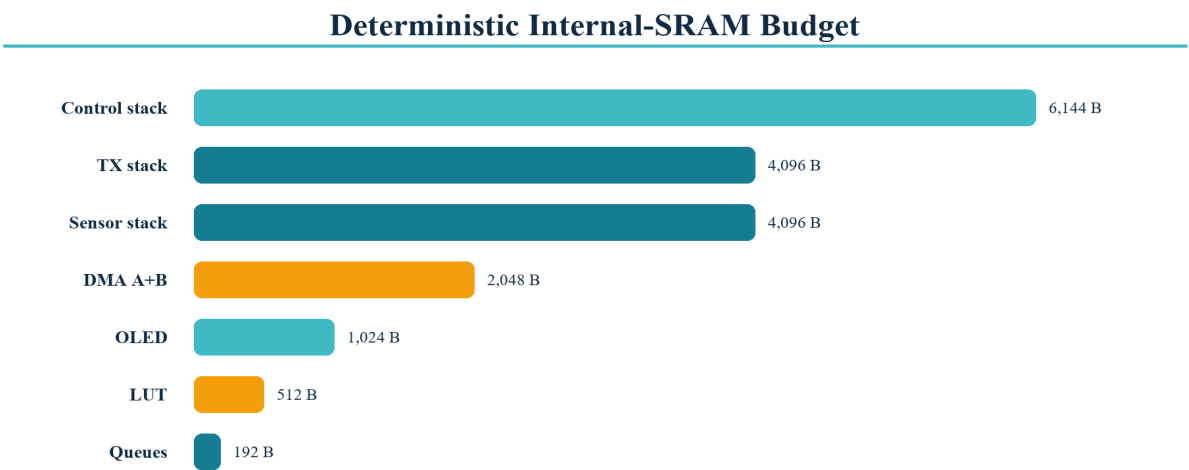


### Safe Transmit and Mission-Mode State Machine



A 1 MHz GPTimer supplies the deterministic start edge. GPIO 3 marks the complete transmit interval; GPIO 2 exposes the Core 0 idle hook. Driver enable is low in boot, ready and fault states.

### 3. Memory and signal generation



The Q16.16 phase accumulator indexes a 256-entry Q15 sine table. Rectangular, Hann, Tukey and Blackman envelopes use integer arithmetic. The normal sample path contains no floating-point operation. A compile flag enables the naive `sinf()` loop only for controlled A/B testing.

| Mode     | Window   | Samples | Duration | Rate   |
|----------|----------|---------|----------|--------|
| Range    | Tukey    | 4,096   | 2.048 ms | 2 MSps |
| Balanced | Hann     | 3,072   | 1.536 ms | 2 MSps |
| Detail   | Blackman | 2,048   | 1.024 ms | 2 MSps |

## 4. BLE contract

Service UUID: 47524e53-1241-659e-584b-90b18f052601. Command UUID ends in 2602; telemetry UUID ends in 2603. Packets are little-endian and protected by CRC-16/CCITT-FALSE.

| Packet    | Bytes | Direction      | Required fields                                                    |
|-----------|-------|----------------|--------------------------------------------------------------------|
| Command   | 20    | App to payload | Magic, version, opcode, sequence, two arguments, CRC               |
| Telemetry | 48    | Payload to app | State, plan, water, battery, current, energy, latency, faults, CRC |

## Mission guard

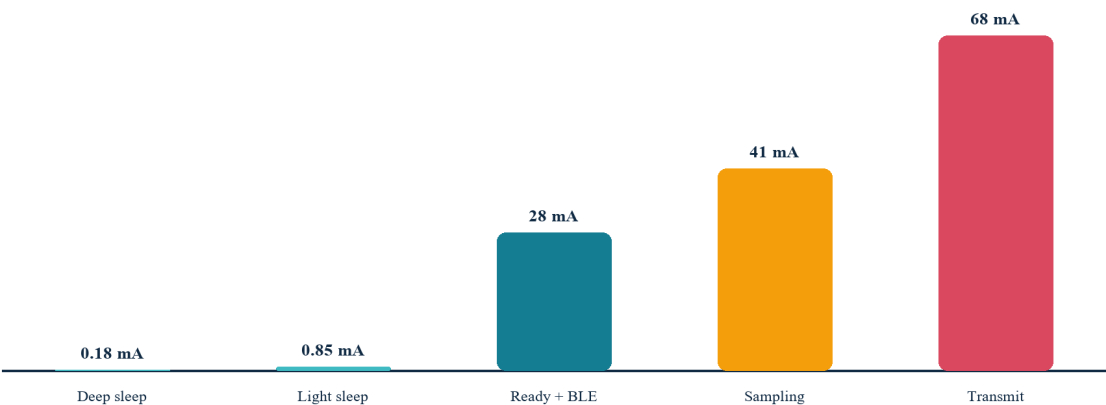
Opcode 3 requires argument 0 equal to 0x534E5247, the ASCII word SNRG, in addition to a valid packet CRC. Mission entry disables BLE. The local trigger exits mission mode.

## Telemetry scaling

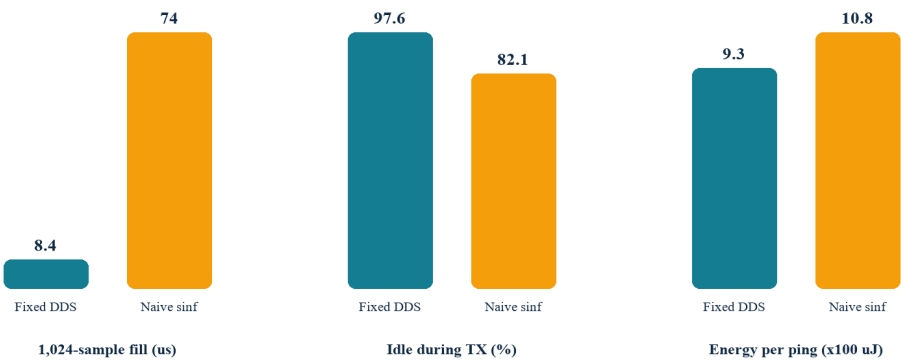
| Field              | Scaling                     |
|--------------------|-----------------------------|
| Temperature        | 0.01 deg C                  |
| Turbidity          | 0.1 NTU                     |
| Battery            | mV                          |
| Current            | 0.1 mA                      |
| Energy             | microjoules                 |
| CPU idle           | permille; 65535 unavailable |
| Adaptation latency | ms                          |

# 5. Power and verification

Power-State Engineering Baselines



DDS A/B Engineering Baseline



Engineering baselines provide instrument ranges and acceptance thresholds. Final evidence is produced on the same board using INA226 integration and two-channel scope captures. The inherited sensor session records 28.69 to 28.75 deg C and stable ADS1115 channels around 0.57 V.

## 6. Build and release

Build from 02-firmware/src using ESP-IDF: set target esp32s3, run idf.py build, then flash and monitor the board. The offline bundle uses bootloader at 0x0000, partition table at 0x8000 and application at 0x10000.

| Release gate        | Evidence                                    |
|---------------------|---------------------------------------------|
| Normal DDS build    | Naive-trig flag disabled                    |
| Internal DMA memory | idf.py size-components and map file         |
| Protocol            | Command and telemetry packet vectors        |
| Timing              | GPIO 2/GPIO 3 scope capture                 |
| Power               | INA226 state CSV and 30-ping statistics     |
| Identity            | Version, commit, board revision and SHA-256 |

This package defines the complete implementation and verification contract. A binary is added to releases only after it is generated by the documented ESP-IDF build and tied to a source revision.